Trabalho Final — Mininet

Questão 1 — Topologia Linear com 8 Switches

a) Criação da topologia

Topologia linear com 8 switches criada com MACs padronizados e largura de banda de 20 Mbps:

```
sudo mn --topo linear,8 --mac --link tc,bw=20
```

```
mininet@mininet-vm:~$ sudo mn --topo linear,8 --mac --link tc,bw=20
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8
*** Adding links:
(20.00Mbit) (20.00Mbit) (h1, s1) (20.00Mbit) (20.00Mbit) (h2, s2) (20.00Mbit) (2
0.00Mbit) (h3, s3) (20.00Mbit) (20.00Mbit) (h4, s4) (20.00Mbit) (20.00Mbit) (h5,
 s5) (20.00Mbit) (20.00Mbit) (h6, s6) (20.00Mbit) (20.00Mbit) (h7, s7) (20.00Mbi
t) (20.00Mbit) (h8, s8) (20.00Mbit) (20.00Mbit) (s2, s1) (20.00Mbit) (20.00Mbit)
 (s3, s2) (20.00Mbit) (20.00Mbit) (s4, s3) (20.00Mbit) (20.00Mbit) (s5, s4) (20.
00Mbit) (20.00Mbit) (s6, s5) (20.00Mbit) (20.00Mbit) (s7, s6) (20.00Mbit) (20.00
Mbit) (s8, s7)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller
c0
*** Starting 8 switches
s1 s2 s3 s4 s5 s6 s7 s8 ...(20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.0
0Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (
20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbi
t) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit)
*** Starting CLI:
mininet>
```

b) Inspeção de interfaces, MACs, IPs e portas

net e dump

O comando net exibe interfaces e links. O dump mostra IP, interface e PID de cada nó:

```
mininet> net
mininet> dump
```

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
h3 h3-eth0:s3-eth1
h4 h4-eth0:s4-eth1
h5 h5-eth0:s5-eth1
h6 h6-eth0:s6-eth1
h7 h7-eth0:s7-eth1
h8 h8-eth0:s8-eth1
s1 lo:  s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo:  s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2
s3 lo:  s3-eth1:h3-eth0 s3-eth2:s2-eth3 s3-eth3:s4-eth2
s4 lo:  s4-eth1:h4-eth0 s4-eth2:s3-eth3 s4-eth3:s5-eth2
s5 lo:  s5-eth1:h5-eth0 s5-eth2:s4-eth3 s5-eth3:s6-eth2
s6 lo:  s6-eth1:h6-eth0 s6-eth2:s5-eth3 s6-eth3:s7-eth2
s7 lo:  s7-eth1:h7-eth0 s7-eth2:s6-eth3 s7-eth3:s8-eth2
s8 lo:  s8-eth1:h8-eth0 s8-eth2:s7-eth3
c0
mininet>
```

```
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1710>
<Host h2: h2-eth0:10.0.0.2 pid=1712>
<Host h3: h3-eth0:10.0.0.3 pid=1714>
<Host h4: h4-eth0:10.0.0.4 pid=1716>
<Host h5: h5-eth0:10.0.0.5 pid=1718>
<Host h6: h6-eth0:10.0.0.6 pid=1720>
<Host h7: h7-eth0:10.0.0.7 pid=1722>
<Host h8: h8-eth0:10.0.0.8 pid=1724>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=1729>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=1732>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=1735>
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=1738>
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=1741>
<OVSSwitch s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None pid=1744>
<OVSSwitch s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None pid=1747>
<OVSSwitch s8: lo:127.0.0.1,s8-eth1:None,s8-eth2:None pid=1750>
<Controller c0: 127.0.0.1:6653 pid=1703>
mininet>
```

ifconfig — h1, h3, h5 e h8

Endereço IP e MAC de cada host:

```
mininet> h1 ifconfig
```

```
mininet> h1 ifconfig
h1-eth0   Link encap:Ethernet  HWaddr 00:00:00:00:00:01
          inet addr:10.0.0.1  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet>
```

```
mininet> h3 ifconfig
```

```
mininet> h3 ifconfig
h3-eth0   Link encap:Ethernet  HWaddr 00:00:00:00:00:03
          inet addr:10.0.0.3  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet>
```

```
mininet> h5 ifconfig
```

```
mininet> h5 ifconfig
h5-eth0   Link encap:Ethernet  HWaddr 00:00:00:00:00:05
          inet addr:10.0.0.5  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet>
```

```
mininet> h8 ifconfig
```

```
mininet> h8 ifconfig
h8-eth0   Link encap:Ethernet  HWaddr 00:00:00:00:00:08
          inet addr:10.0.0.8  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet>
```

Portas dos switches — ovs-vsctl show

O ovs-vsctl show exibe as bridges OVS com portas e controlador conectado:

```
mininet> s1 ovs-ofctl show s1
mininet> s1 ovs-vsctl show
```

```
mininet> s1 ovs-ofctl show s1
ovs-ofctl: 127.0.0.1 is not a bridge or a socket
mininet> s2 ovs-ofctl show s2
ovs-ofctl: 127.0.0.1 is not a bridge or a socket
mininet> s8 ovs-ofctl show s8
ovs-ofctl: 127.0.0.1 is not a bridge or a socket
mininet>
```

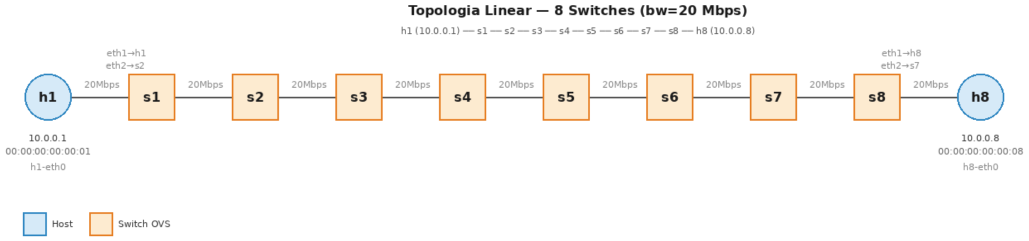
```
mininet> s1 ovs-vsctl show
918037ec-b307-45d7-a75a-f1ac4337d135
    Bridge "s7"
        Controller "tcp:127.0.0.1:6653"
            is_connected: true
        Controller "ptcp:6660"
        fail_mode: secure
        Port "s7-eth2"
            Interface "s7-eth2"
        Port "s7-eth1"
            Interface "s7-eth1"
        Port "s7"
            Interface "s7"
                type: internal
        Port "s7-eth3"
            Interface "s7-eth3"
    Bridge "s5"
        Controller "tcp:127.0.0.1:6653"
            is_connected: true
        Controller "ptcp:6658"
        fail_mode: secure
        Port "s5-eth3"
            Interface "s5-eth3"
        Port "s5"
            Interface "s5"
                type: internal
        Port "s5-eth2"
            Interface "s5-eth2"
        Port "s5-eth1"
            Interface "s5-eth1"
    Bridge "s4"
        Controller "tcp:127.0.0.1:6653"
            is_connected: true
        Controller "ptcp:6657"
        fail_mode: secure
        Port "s4-eth3"
            Interface "s4-eth3"
        Port "s4"
            Interface "s4"
                type: internal
        Port "s4-eth1"
            Interface "s4-eth1"
        Port "s4-eth2"
            Interface "s4-eth2"
```

```
Bridge "s2"
    Controller "tcp:127.0.0.1:6653"
        is_connected: true
    Controller "ptcp:6655"
    fail_mode: secure
    Port "s2-eth3"
        Interface "s2-eth3"
    Port "s2-eth2"
        Interface "s2-eth2"
    Port "s2"
        Interface "s2"
            type: internal
    Port "s2-eth1"
        Interface "s2-eth1"
Bridge "s8"
    Controller "ptcp:6661"
    Controller "tcp:127.0.0.1:6653"
        is_connected: true
    fail_mode: secure
    Port "s8-eth1"
        Interface "s8-eth1"
    Port "s8-eth2"
        Interface "s8-eth2"
    Port "s8"
        Interface "s8"
            type: internal
Bridge "s1"
    Controller "tcp:127.0.0.1:6653"
        is_connected: true
    Controller "ptcp:6654"
    fail_mode: secure
    Port "s1-eth1"
        Interface "s1-eth1"
    Port "s1-eth2"
        Interface "s1-eth2"
    Port "s1"
        Interface "s1"
            type: internal
```

```
    Bridge "s3"
        Controller "ptcp:6656"
        Controller "tcp:127.0.0.1:6653"
            is_connected: true
        fail_mode: secure
        Port "s3-eth3"
            Interface "s3-eth3"
        Port "s3"
            Interface "s3"
                type: internal
        Port "s3-eth1"
            Interface "s3-eth1"
        Port "s3-eth2"
            Interface "s3-eth2"
    Bridge "s6"
        Controller "ptcp:6659"
        Controller "tcp:127.0.0.1:6653"
            is_connected: true
        fail_mode: secure
        Port "s6-eth1"
            Interface "s6-eth1"
        Port "s6"
            Interface "s6"
                type: internal
        Port "s6-eth3"
            Interface "s6-eth3"
        Port "s6-eth2"
            Interface "s6-eth2"
    ovs_version: "2.0.2"
mininet>
```

c) Desenho ilustrativo da topologia

Topologia linear com 8 switches em sequência. Todos os links operam a 20 Mbps:



d) Testes de ping e tcpdump

pingall

Resultado: 0% de perda (56/56 recebidos):

```
mininet> pingall
```

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
mininet>
```

tcpdump — h1 para h3

O tcpdump rodou em background no h3 enquanto o ping era disparado do h1, mostrando os pacotes ICMP chegando:

```
mininet> h3 tcpdump -i h3-eth0 icmp -n &
mininet> h1 ping -c 5 h3
```

```
mininet> h3 tcpdump -i h3-eth0 icmp -n &
mininet> h1 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=7.22 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.292 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.056 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.051 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.053 ms

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.051/1.534/7.222/2.845 ms
mininet>
```

e) Testes iperf — servidor TCP na porta 5555

O h1 foi configurado como servidor TCP (porta 5555) e h3 como cliente. Teste de 15 segundos com relatório a cada 1 segundo para bw=20, 30 e 40 Mbps.

bw = 20 Mbps — banda medida: ~19,3 Mbits/s

```
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
```

```
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on h3-eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  3] local 10.0.0.3 port 58518 connected with 10.0.0.1 port 5555
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0- 1.0 sec  2.62 MBytes  22.0 Mbits/sec
[  3]  1.0- 2.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  2.0- 3.0 sec  2.38 MBytes  19.9 Mbits/sec
[  3]  3.0- 4.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  4.0- 5.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  5.0- 6.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  6.0- 7.0 sec  2.38 MBytes  19.9 Mbits/sec
[  3]  7.0- 8.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  8.0- 9.0 sec  2.38 MBytes  19.9 Mbits/sec
[  3]  9.0-10.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3] 10.0-11.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3] 11.0-12.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3] 12.0-13.0 sec  2.38 MBytes  19.9 Mbits/sec
[  3] 13.0-14.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3] 14.0-15.0 sec  2.25 MBytes  18.9 Mbits/sec
[  3]  0.0-15.1 sec  34.8 MBytes  19.3 Mbits/sec
mininet> 
```

bw = 30 Mbps — banda medida: ~28,9 Mbits/s

```
sudo mn --topo linear,8 --mac --link tc,bw=30
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
```

```
mininet@mininet-vm:~$ sudo mn --topo linear,8 --mac --link tc,bw=30
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8
*** Adding links:
(30.00Mbit) (30.00Mbit) (h1, s1) (30.00Mbit) (30.00Mbit) (h2, s2) (30.00Mbit) (3
0.00Mbit) (h3, s3) (30.00Mbit) (30.00Mbit) (h4, s4) (30.00Mbit) (30.00Mbit) (h5,
 s5) (30.00Mbit) (30.00Mbit) (h6, s6) (30.00Mbit) (30.00Mbit) (h7, s7) (30.00Mbi
t) (30.00Mbit) (h8, s8) (30.00Mbit) (30.00Mbit) (s2, s1) (30.00Mbit) (30.00Mbit)
 (s3, s2) (30.00Mbit) (30.00Mbit) (s4, s3) (30.00Mbit) (30.00Mbit) (s5, s4) (30.
00Mbit) (30.00Mbit) (s6, s5) (30.00Mbit) (30.00Mbit) (s7, s6) (30.00Mbit) (30.00
Mbit) (s8, s7)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller
c0
*** Starting 8 switches
s1 s2 s3 s4 s5 s6 s7 s8 ...(30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.0
0Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (
30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbi
t) (30.00Mbit) (30.00Mbit) (30.00Mbit) (30.00Mbit)
*** Starting CLI:
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  3] local 10.0.0.3 port 58554 connected with 10.0.0.1 port 5555
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0- 1.0 sec  3.88 MBytes  32.5 Mbits/sec
[  3]  1.0- 2.0 sec  3.50 MBytes  29.4 Mbits/sec
[  3]  2.0- 3.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  3.0- 4.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  4.0- 5.0 sec  3.50 MBytes  29.4 Mbits/sec
[  3]  5.0- 6.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  6.0- 7.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  7.0- 8.0 sec  3.50 MBytes  29.4 Mbits/sec
[  3]  8.0- 9.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  9.0-10.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3] 10.0-11.0 sec  3.50 MBytes  29.4 Mbits/sec
[  3] 11.0-12.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3] 12.0-13.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3] 13.0-14.0 sec  3.50 MBytes  29.4 Mbits/sec
[  3] 14.0-15.0 sec  3.38 MBytes  28.3 Mbits/sec
[  3]  0.0-15.1 sec  51.9 MBytes  28.9 Mbits/sec
mininet>
```

bw = 40 Mbps — banda medida: ~38,5 Mbits/s

```
sudo mn --topo linear,8 --mac --link tc,bw=40
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
```

```
mininet@mininet-vm:~$ sudo mn --topo linear,8 --mac --link tc,bw=40
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8
*** Adding links:
(40.00Mbit) (40.00Mbit) (h1, s1) (40.00Mbit) (40.00Mbit) (h2, s2) (40.00Mbit) (4
0.00Mbit) (h3, s3) (40.00Mbit) (40.00Mbit) (h4, s4) (40.00Mbit) (40.00Mbit) (h5,
 s5) (40.00Mbit) (40.00Mbit) (h6, s6) (40.00Mbit) (40.00Mbit) (h7, s7) (40.00Mbi
t) (40.00Mbit) (h8, s8) (40.00Mbit) (40.00Mbit) (s2, s1) (40.00Mbit) (40.00Mbit)
 (s3, s2) (40.00Mbit) (40.00Mbit) (s4, s3) (40.00Mbit) (40.00Mbit) (s5, s4) (40.
00Mbit) (40.00Mbit) (s6, s5) (40.00Mbit) (40.00Mbit) (s7, s6) (40.00Mbit) (40.00
Mbit) (s8, s7)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller
c0
*** Starting 8 switches
s1 s2 s3 s4 s5 s6 s7 s8 ...(40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.0
0Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (
40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbi
t) (40.00Mbit) (40.00Mbit) (40.00Mbit) (40.00Mbit)
*** Starting CLI:
mininet> h1 iperf -s -p 5555 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -t 15 -i 1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  3] local 10.0.0.3 port 58574 connected with 10.0.0.1 port 5555
[ ID] Interval       Transfer     Bandwidth
[  3]  0.0- 1.0 sec  5.12 MBytes  43.0 Mbits/sec
[  3]  1.0- 2.0 sec  4.75 MBytes  39.8 Mbits/sec
[  3]  2.0- 3.0 sec  4.38 MBytes  36.7 Mbits/sec
[  3]  3.0- 4.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3]  4.0- 5.0 sec  4.75 MBytes  39.8 Mbits/sec
[  3]  5.0- 6.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3]  6.0- 7.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3]  7.0- 8.0 sec  4.38 MBytes  36.7 Mbits/sec
[  3]  8.0- 9.0 sec  4.75 MBytes  39.8 Mbits/sec
[  3]  9.0-10.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3] 10.0-11.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3] 11.0-12.0 sec  4.75 MBytes  39.8 Mbits/sec
[  3] 12.0-13.0 sec  4.38 MBytes  36.7 Mbits/sec
[  3] 13.0-14.0 sec  4.50 MBytes  37.7 Mbits/sec
[  3] 14.0-15.0 sec  4.75 MBytes  39.8 Mbits/sec
[  3]  0.0-15.0 sec  69.1 MBytes  38.5 Mbits/sec
mininet> 
```

A banda medida ficou ligeiramente abaixo do limite configurado em todos os testes, o que é esperado pela passagem por 8 switches em série.

Questão 2 — Topologia Customizada em Python

a) Criação da topologia customizada

A topologia foi criada via código Python com MACs padronizados e controlador manual (RemoteController na porta 6653). O switch s1 conecta s2, s3 e s7. Os hosts h1 e h2 estão em s2, h3 em s3, e h4, h5 e h6 em s7.

```
sudo python3 topologia_customizada.py
```

b) Inspeção de interfaces, MACs, IPs e portas

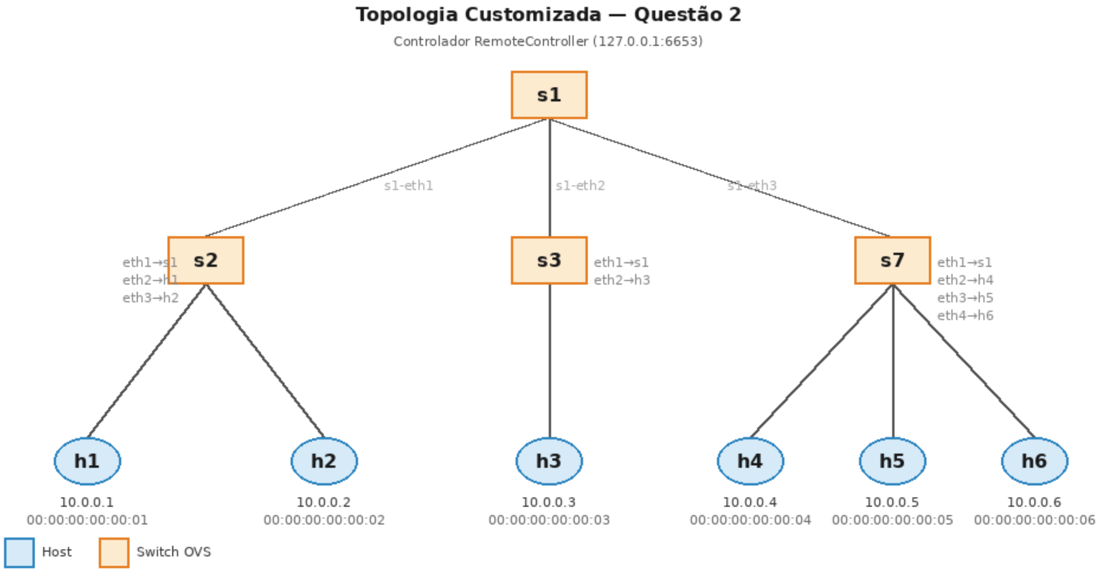
Comandos net, dump e nodes executados para inspecionar todos os nós, interfaces e endereços:

```
mininet> net
mininet> dump
mininet> nodes
```

```
*** Starting CLI:
mininet> net
h1 h1-eth0:s2-eth2
h2 h2-eth0:s2-eth3
h3 h3-eth0:s3-eth2
h4 h4-eth0:s7-eth2
h5 h5-eth0:s7-eth3
h6 h6-eth0:s7-eth4
s1 lo:  s1-eth1:s2-eth1 s1-eth2:s3-eth1 s1-eth3:s7-eth1
s2 lo:  s2-eth1:s1-eth1 s2-eth2:h1-eth0 s2-eth3:h2-eth0
s3 lo:  s3-eth1:s1-eth2 s3-eth2:h3-eth0
s7 lo:  s7-eth1:s1-eth3 s7-eth2:h4-eth0 s7-eth3:h5-eth0 s7-eth4:h6-eth0
c0
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=2184>
<Host h2: h2-eth0:10.0.0.2 pid=2186>
<Host h3: h3-eth0:10.0.0.3 pid=2188>
<Host h4: h4-eth0:10.0.0.4 pid=2190>
<Host h5: h5-eth0:10.0.0.5 pid=2192>
<Host h6: h6-eth0:10.0.0.6 pid=2194>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=2170>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=2173>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=2176>
<OVSSwitch s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None,s7-eth4:None pid=2179>
<RemoteController c0: 127.0.0.1:6653 pid=2163>
mininet> nodes
available nodes are:
c0 h1 h2 h3 h4 h5 h6 s1 s2 s3 s7
mininet>
```

c) Desenho ilustrativo da topologia

Topologia em árvore com s1 como raiz, conectando s2, s3 e s7. Os hosts estão nas folhas de cada switch:



d) Testes de ping com switches normais

Com o controlador em modo learning, o pingall testou a conectividade entre todos os pares. Resultado: 0% de perda (30/30 recebidos):

```
mininet> pingall
```

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6
h2 -> h1 h3 h4 h5 h6
h3 -> h1 h2 h4 h5 h6
h4 -> h1 h2 h3 h5 h6
h5 -> h1 h2 h3 h4 h6
h6 -> h1 h2 h3 h4 h5
*** Results: 0% dropped (30/30 received)
mininet>
```

e) Regras baseadas em endereços MAC

As regras anteriores foram apagadas e novas regras OpenFlow foram instaladas via script bash, associando pares de hosts por MAC. Regras ARP foram adicionadas para resolução de endereços. Ao final, uma regra de drop de prioridade baixa bloqueia o restante do tráfego:

```
sudo bash regras_mac.sh
```

Passo 1 — apagar regras e verificar portas de s1 e s2:

```
mininet@mininet-vm:~$ sudo bash regras_mac.sh
=== PASSO 1: Apagar TODAS as regras existentes ===
Regras apagadas.

=== PASSO 2: Verificar portas de cada switch ===
--- s1 ---
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000001
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: OUTPUT SET_VLAN_VID SET_VLAN_PCP STRIP_VLAN SET_DL_SRC SET_DL_DST SET_N
W_SRC SET_NW_DST SET_NW_TOS SET_TP_SRC SET_TP_DST ENQUEUE
 1(s1-eth1): addr:02:d6:6f:0d:d6:a7
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s1-eth2): addr:a2:29:58:39:b1:08
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 3(s1-eth3): addr:12:b5:6b:49:9e:e0
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s1): addr:9e:54:aa:53:f2:41
     config:     0
     state:      0
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0
--- s2 ---
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000002
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: OUTPUT SET_VLAN_VID SET_VLAN_PCP STRIP_VLAN SET_DL_SRC SET_DL_DST SET_N
W_SRC SET_NW_DST SET_NW_TOS SET_TP_SRC SET_TP_DST ENQUEUE
 1(s2-eth1): addr:22:6c:49:a8:fe:92
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s2-eth2): addr:aa:5d:c8:ef:97:4d
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 3(s2-eth3): addr:92:4d:56:31:f5:10
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s2): addr:ae:b6:6f:ad:25:46
```

Verificação de portas de s3 e s7:

```
     config:     0
     state:      0
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0
--- s3 ---
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000003
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: OUTPUT SET_VLAN_VID SET_VLAN_PCP STRIP_VLAN SET_DL_SRC SET_DL_DST SET_N
W_SRC SET_NW_DST SET_NW_TOS SET_TP_SRC SET_TP_DST ENQUEUE
 1(s3-eth1): addr:e2:42:46:97:98:34
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s3-eth2): addr:b6:76:40:cc:86:d7
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s3): addr:56:ad:f4:73:82:4f
     config:     0
     state:      0
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0
--- s7 ---
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000007
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: OUTPUT SET_VLAN_VID SET_VLAN_PCP STRIP_VLAN SET_DL_SRC SET_DL_DST SET_N
W_SRC SET_NW_DST SET_NW_TOS SET_TP_SRC SET_TP_DST ENQUEUE
 1(s7-eth1): addr:aa:23:fd:68:c4:ad
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s7-eth2): addr:7e:a5:b1:21:d3:39
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 3(s7-eth3): addr:06:1e:7a:da:27:bf
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 4(s7-eth4): addr:b2:68:8e:02:6b:59
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s7): addr:9e:b3:15:e8:3f:4e
     config:     0
     state:      0
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0
```

Instalação das regras MAC e verificação dos flows em todos os switches:

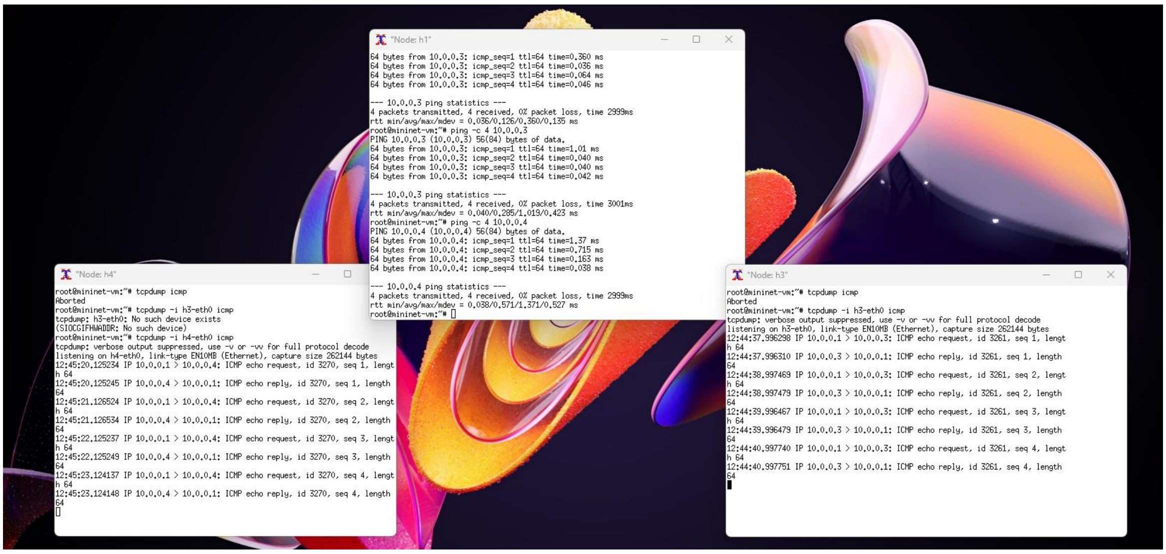
```
=== Instalando regras para permitir tráfego ARP ===

=== PASSO 3: Instalar regras baseadas em MAC ===

=== PASSO 4: Verificar regras instaladas ===
--- Flows em s1 ---
NXST_FLOW reply (xid=0x4):
 cookie=0x0, duration=0.005s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:06 actions=output:3
 cookie=0x0, duration=0.008s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:05 actions=output:3
 cookie=0x0, duration=0.019s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:02 actions=output:1
 cookie=0x0, duration=0.015s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:03 actions=output:2
 cookie=0x0, duration=0.023s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:01 actions=output:1
 cookie=0x0, duration=0.012s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_dst=00:00:00:00:00:04 actions=output:3
 cookie=0x0, duration=0.088s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=50,arp actions=FLOOD
--- Flows em s2 ---
NXST_FLOW reply (xid=0x4):
 cookie=0x0, duration=0.064s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:02 actions=output:3
 cookie=0x0, duration=0.068s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:04 actions=output:1
 cookie=0x0, duration=0.071s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:01 actions=output:2
 cookie=0x0, duration=0.075s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:1
 cookie=0x0, duration=0.087s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=50,arp actions=FLOOD
--- Flows em s3 ---
NXST_FLOW reply (xid=0x4):
 cookie=0x0, duration=0.055s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:05,dl_dst=00:00:00:00:00:03 actions=output:2
 cookie=0x0, duration=0.059s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:05 actions=output:1
 cookie=0x0, duration=0.067s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:01 actions=output:1
 cookie=0x0, duration=0.063s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:2
 cookie=0x0, duration=0.089s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=50,arp actions=FLOOD
--- Flows em s7 ---
NXST_FLOW reply (xid=0x4):
 cookie=0x0, duration=0.047s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:05,dl_dst=00:00:00:00:00:03 actions=output:1
 cookie=0x0, duration=0.043s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:05 actions=output:3
 cookie=0x0, duration=0.057s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:02 actions=output:1
 cookie=0x0, duration=0.051s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=100,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:04 actions=output:2
 cookie=0x0, duration=0.091s, table=0, n_packets=0, n_bytes=0, idle_age=0, prior
ity=50,arp actions=FLOOD
```

Comandos de teste executados dentro do Mininet CLI:

```
=== PASSO 5: Testar conectividade entre switches diferentes ===
Para testar dentro do Mininet CLI:
  mininet> h1 ping -c 3 h3
  mininet> h2 ping -c 3 h4
  mininet> h3 ping -c 3 h5
  mininet> h1 ping -c 3 h5
```

f) Testes de ping com regras MAC

Os testes foram realizados com xterm abrindo terminais individuais. Os pings entre h1↔h3, h4↔h1 e h5↔h3 funcionam corretamente — todos hosts de switches diferentes — confirmando que as regras MAC estão ativas:



Teste de bloqueio: h1 tentando pingar h4 após aplicação da regra drop. Resultado: 100% de perda de pacotes, confirmando que o tráfego não autorizado pelas regras MAC é descartado:

```
ping -c 4 10.0.0.4
```

```
root@mininet-vm:~# ping -c 4 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3023ms

root@mininet-vm:~#
```